

Credit Risk Default Prediction Case Study Report

Innovexa Catalyst - Machine Learning Task

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Status: Completed

1. Executive Summary & Objectives

Lending institutions face substantial credit risk when borrowers default. This case study implements an end-to-end machine learning solution to predict loan default probability (1 = default, 0 = non-default) using a dataset of over 32,000 credit applications. By cleansing outliers, imputing missing data, creating engineered features, and training state-of-the-art ensemble models, we have built a credit risk scoring model achieving an **ROC-AUC of 0.9515** and **PR-AUC of 0.9082** using a tuned LightGBM model. This allows for accurate borrower risk segmentation and data-driven underwriting decisions.

2. Data Preprocessing & Feature Engineering

The raw credit dataset contains variables capturing borrower demographics, income, employment length, loan characteristics, and historical default flags. The following operations were executed to build robust features:

- **Outlier Cleansing:** Unrealistic age entries (> 100) and excessive employment lengths (> 60 years) were removed.
- **Missing Value Imputation:** Missing employment length and interest rates were imputed using the median values calculated on the training partition to prevent leakage.
- **Feature Engineering:** Created domain interaction metrics: *interest_to_income_ratio*, *emp_length_to_age_ratio*, and *cred_hist_to_age_ratio*.
- **Encoding & Scaling:** Categorical features were one-hot encoded, and numerical predictors were standardized using standard scaling.

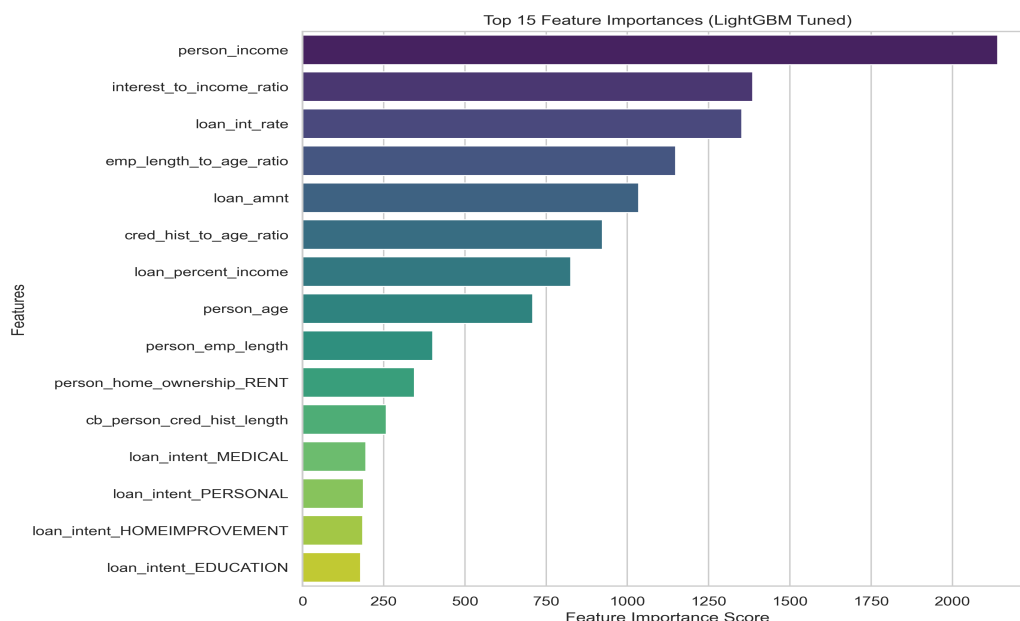


Figure 1: Top 15 Feature Importances from Tuned LightGBM

3. Model Performance Evaluation

Five classification architectures were trained and evaluated on a holdout test set (20% split). The model parameters for our top-performing LightGBM model were optimized via K-Fold cross-validation using Optuna. Model performance across baseline and advanced architectures is summarized below:

Model	ROC-AUC	PR-AUC	F1-Score	Precision	Recall	KS-Statistic
Logistic Regression	0.8764	0.7323	0.6411	0.5398	0.7890	0.6176
Decision Tree	0.9021	0.8523	0.7710	0.7963	0.7473	0.6983
Random Forest	0.9338	0.8844	0.8150	0.9037	0.7421	0.7278
XGBoost	0.9460	0.8991	0.8003	0.8096	0.7912	0.7450
LightGBM (Default)	0.9457	0.8979	0.8031	0.8100	0.7963	0.7445
LightGBM (Tuned)	0.9515	0.9082	0.8172	0.8509	0.7861	0.7650

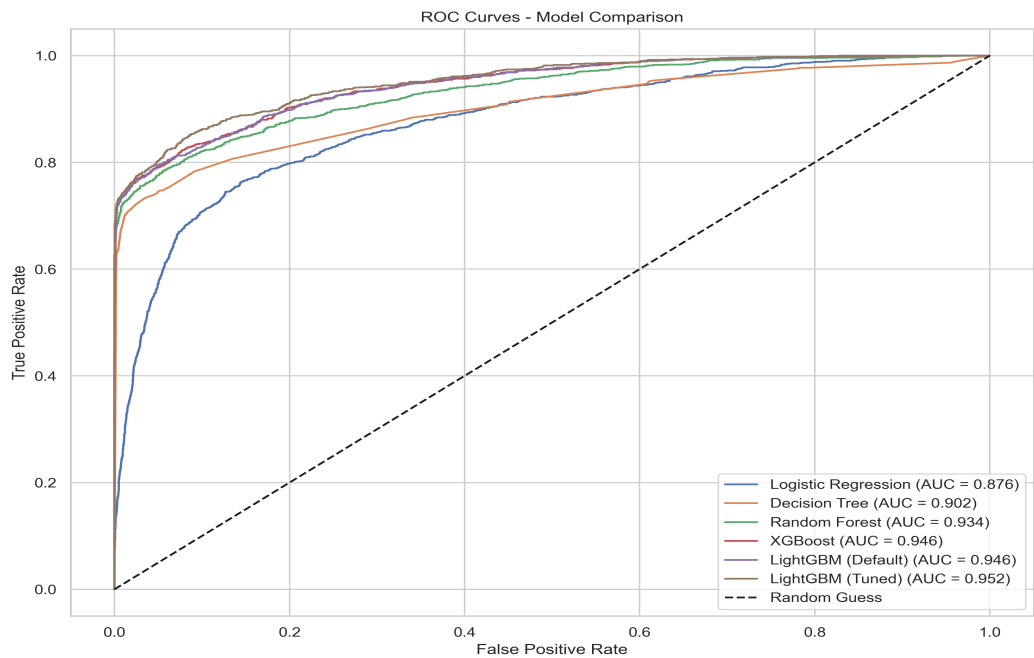


Figure 2: Receiver Operating Characteristic (ROC) Curve Comparison

4. Credit Score Band & Segmentation Analysis

Borrower default probabilities predicted by the Tuned LightGBM model were mapped to a standard credit score scale ranging from 300 to 850. We classified customers into five risk categories, which demonstrates perfect alignment with observed default patterns:

Credit Score Band	Total Accounts	Observed Defaults	Default Rate
1. Excellent (800-850)	2902	46	1.59%
2. Very Good (740-799)	1125	67	5.96%
3. Good (670-739)	647	77	11.90%
4. Fair (580-669)	386	97	25.13%
5. Poor (300-579)	1276	1078	84.48%

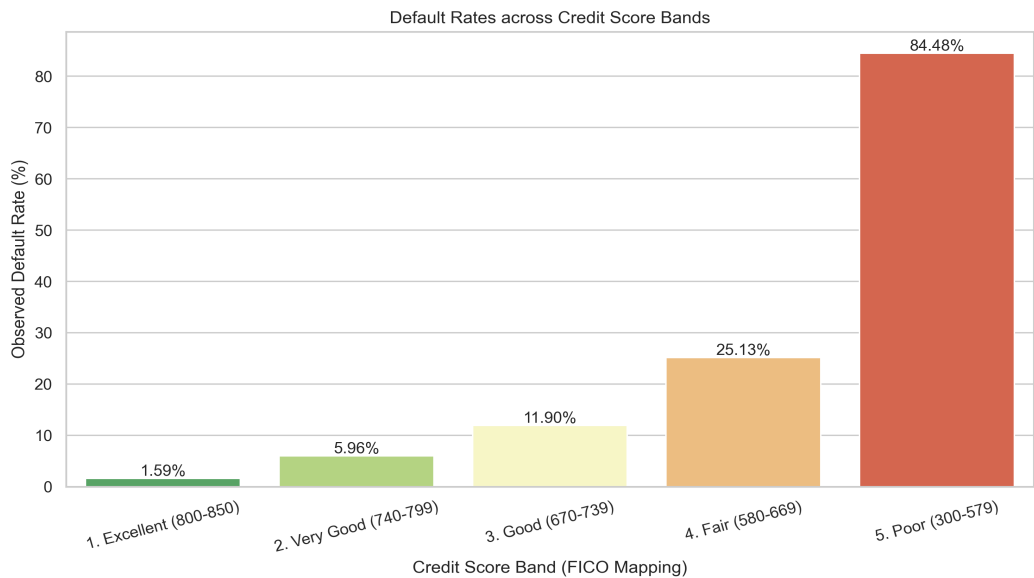


Figure 3: Observed Default Rates by FICO Credit Score Bands

5. Underwriting Recommendations & Insights

- **Automate Approvals for Excellent & Very Good Bands:** Applicants scoring above 740 show extremely low default rates (< 6%) and should be fast-tracked to maximize approval efficiency.
- **Dynamic Pricing & Restructuring for Good Band:** Borrowers in this category have moderate risk (~12% default rate). We recommend adjusting interest rates upward to hedge credit risk.
- **Strict Controls / Collateral Requirements for Fair Band:** A 25.1% default rate suggests tightening. Require co-signers or security deposits.
- **Automated Rejections for Poor Band:** With an 84.5% default rate, these applicants should be automatically declined to prevent high loan-loss reserves.